where $s_4 = \sin x_4$ and $c_4 = \cos x_4$. The linear system

$$\dot{\tilde{x}} = A\tilde{x} + B\tilde{u} \tag{1.99}$$

approximately describes the deviation of x from its nominal value $\bar{x}$. The nonlinear system was simulated with the nominal control values $\bar{u}_a(t) = \sin 2\pi t$ and $\bar{u}_b(t) = \cos 2\pi t$. This resulted in a nominal state trajectory $\bar{x}(t)$. The linear and nonlinear systems were then simulated with nonnominal control values. Figure 1.1 shows the results of the linear and nonlinear simulations when the control magnitude deviation from nominal is a small positive number. It can be seen that the simulations result in similar state-space trajectories, although they do not match exactly. If the deviation is zero, then the linear and nonlinear simulations will match exactly. As the deviation from nominal increases, the difference between the linear and nonlinear simulations will increase.

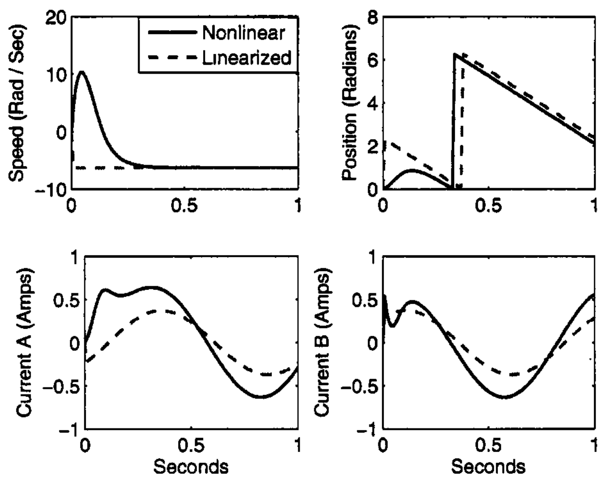

Figure 1.1 Example 1.4 comparison of nonlinear and linearized motor simulations.

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1.4 DISCRETIZATION

Most systems in the real world are described with continuous-time dynamics of the type shown in Equations (1.67) or (1.83). However, state estimation and control algorithms are almost always implemented in digital electronics. This often requires a transformation of continuous-time dynamics to discrete-time dynamics. This section discusses how a continuous-time linear system can be transformed into a discrete-time linear system.

Recall from Equation (1.68) that the solution of a continuous-time linear system is given by

$$x(t) = e^{A(t-t_0)}x(t_0) + \int_{t_0}^{t} e^{A(t-\tau)}Bu(\tau)\, d\tau \tag{1.100}$$